Radian Measure and Arc Length

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Converting between degrees and radians**

Convert each angle to radians, in exact form:

a $45° = \frac{\pi}{4}$ **b** $120° = \frac{2\pi}{3}$ **c** $270° = \frac{3\pi}{2}$ **d** $330° = \frac{11\pi}{6}$

(a) $= \frac{\pi}{4}$; (b) $= \frac{2\pi}{3}$; (c) $= \frac{3\pi}{2}$; (d) $= \frac{11\pi}{6}$

2 **[Foundational]** • **Target: Converting between degrees and radians**

Convert each angle to degrees. Round to one decimal place where needed:

a $\frac{\pi}{6} = 30°$ **b** $\frac{5\pi}{4} = 225°$ **c** $\frac{7\pi}{3} = 420°$ **d** 3.5 rad $\approx 200.5°$

(a) $= 30°$; (b) $= 225°$; (c) $= 420°$; (d) $\approx 200.5°$

3 **[Foundational]** • **Target: Arc length and sector area**

A circle has radius 6 cm. Find the exact length of the arc subtended by a central angle of $\frac{2\pi}{3}$.

$s = r\theta = 6 \cdot \frac{2\pi}{3} = 4\pi \approx 12.57$ cm.

4 **[Foundational]** • **Target: Arc length and sector area**

A sector of a circle of radius 10 cm has central angle $\frac{\pi}{5}$. Find its exact area.

$A = \frac{1}{2}r^2\theta = \frac{1}{2}(100)\frac{\pi}{5} = 10\pi \approx 31.42$ cm^2.

5 **[Foundational]** • **Target: Arc length and sector area**

A wheel of radius 30 cm rolls along the ground without slipping.

a Through what angle, in radians, does the wheel turn when it travels 1.5 m? $\theta = \frac{s}{r} = \frac{150}{30} = 5$ radians.

b Express this angle to the nearest tenth of a degree. $5 \times \frac{180°}{\pi} \approx 286.5°$.

(a) $\theta = \frac{s}{r} = \frac{150}{30} = 5$ radians.; (b) $5 \times \frac{180°}{\pi} \approx 286.5°$.

6 **[Foundational]** • **Target: Arc length and sector area**

A pendulum of length 80 cm swings through an angle of $25°$. Find the length of the arc traced by its tip, to the nearest tenth of a centimetre.

$\theta = 25° = \frac{25\pi}{180}$ rad, so $s = 80 \cdot \frac{25\pi}{180} \approx 34.9$ cm.

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7 [Foundational] • Target: Angular and linear speed

A carousel horse is 4 m from the center and completes one full rotation every 20 seconds.

a Find the angular speed of the horse, in radians per second. $\omega = \frac{2\pi}{20} = \frac{\pi}{10}$ rad/s.

b Find the linear speed of the horse, in m/s. $v = r\omega = 4 \cdot \frac{\pi}{10} = \frac{2\pi}{5} \approx 1.26$ m/s.

(a) $\omega = \frac{2\pi}{20} = \frac{\pi}{10}$ rad/s.; (b) $v = r\omega = 4 \cdot \frac{\pi}{10} = \frac{2\pi}{5} \approx 1.26$ m/s.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Angular and linear speed

A car's wheel has radius 35 cm and spins at 400 revolutions per minute. Find the car's speed in km/h, to one decimal place.

$v = r\omega$, with $\omega = 400 \times 2\pi$ rad/min $= 800\pi$ rad/min. $v = 0.35 \times 800\pi \approx 879.6$ m/min $= 52{,}776$ m/h ≈ 52.8 km/h.

9 [Intermediate] • Target: Angular and linear speed

Two gears mesh together: the smaller gear has radius 5 cm and spins at 300 rpm. If the larger gear has radius 15 cm, find its rotational speed in rpm (the linear speed at the point of contact is the same for both gears).

$v = r_1\omega_1 = r_2\omega_2$: $5(300) = 15\,\omega_2$, so $\omega_2 = 100$ rpm.

10 [Intermediate] • Target: Coterminal angles and reference angles

Find a positive coterminal angle (between 0 and 2π) for each:

a $\frac{9\pi}{4} = \frac{\pi}{4}$ **b** $-\frac{\pi}{3} = \frac{5\pi}{3}$ **c** $\frac{13\pi}{6} = \frac{\pi}{6}$

(a) $= \frac{\pi}{4}$; (b) $= \frac{5\pi}{3}$; (c) $= \frac{\pi}{6}$

11 [Intermediate] • Target: Coterminal angles and reference angles

Find the reference angle for each angle:

a $\frac{5\pi}{6} = \frac{\pi}{6}$ **b** $\frac{7\pi}{4} = \frac{\pi}{4}$ **c** $\frac{4\pi}{3} = \frac{\pi}{3}$

(a) $= \frac{\pi}{6}$; (b) $= \frac{\pi}{4}$; (c) $= \frac{\pi}{3}$

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12 [Intermediate] • Target: The radian definition itself

State the definition of one radian in terms of arc length and radius, and explain why radians are considered a 'natural' unit for angles.

One radian is the angle subtended at the center of a circle by an arc whose length equals the radius. Radians are natural because the arc length formula $s = r\theta$ has no extra conversion constant when θ is in radians, unlike degrees.

13 [Intermediate] • Target: The radian definition itself

Explain why the full circle corresponds to 2π radians, using the circumference formula $C = 2\pi r$.

Since $s = r\theta$ and the full circumference is $s = 2\pi r$, setting $r\theta = 2\pi r$ gives $\theta = 2\pi$ — the full circle sweeps exactly 2π radians.

14 [Intermediate] • Target: Combined arc length and sector area problems

A sprinkler sprays water over a sector of radius 8 m and central angle $\frac{\pi}{3}$. Find both the area watered and the length of the outer arc of the watered region.

Area: $A = \frac{1}{2}(8)^2\left(\frac{\pi}{3}\right) = \frac{32\pi}{3} \approx 33.5$ m^2. Arc length: $s = 8\left(\frac{\pi}{3}\right) = \frac{8\pi}{3} \approx 8.4$ m.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Combined arc length and sector area problems • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

A slice of pizza (a sector) has radius 12 cm and arc length 10 cm along its crust. Find the central angle in radians and the area of the slice.

$\theta = \frac{s}{r} = \frac{10}{12} = \frac{5}{6}$ rad. Area $= \frac{1}{2}r^2\theta = \frac{1}{2}(144)\left(\frac{5}{6}\right) = 60$ cm^2.

16 [Challenging] • Target: Common angles in both units (reference table) • Cross-Lesson: 7.02 Logarithms and Their Properties

Complete a conversion table for the common angles $0°, 30°, 45°, 60°, 90°, 180°, 360°$ in radians.

$0° = 0$; $30° = \frac{\pi}{6}$; $45° = \frac{\pi}{4}$; $60° = \frac{\pi}{3}$; $90° = \frac{\pi}{2}$; $180° = \pi$; $360° = 2\pi$.

17 [Challenging] • Target: Satellite and orbital speed applications • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

A satellite orbits Earth at a radius of 7000 km, completing one orbit every 2 hours. Find its angular speed in rad/h and its linear speed in km/h, to the nearest whole number.

$\omega = \frac{2\pi}{2} = \pi$ rad/h. $v = r\omega = 7000\pi \approx 21{,}991$ km/h.

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18 [Challenging] • Target: Satellite and orbital speed applications • Cross-Lesson: 7.04 Applications: Growth and Decay

Earth rotates once every 24 hours. Find the linear speed (due to rotation alone) of a point on the equator, using Earth's radius ≈ 6371 km, to the nearest km/h.

$\omega = \frac{2\pi}{24} = \frac{\pi}{12}$ rad/h. $v = 6371 \cdot \frac{\pi}{12} \approx 1668$ km/h.

19 [Challenging] • Target: Negative angles and full-turn arcs • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Find the exact arc length swept by a radius-4 circle for a rotation of $-\frac{3\pi}{2}$ radians (clockwise), and interpret the sign.

Arc length uses the magnitude of θ: $s = 4\left(\frac{3\pi}{2}\right) = 6\pi$. The negative sign in the angle indicates clockwise rotation, but arc length itself is always a positive (unsigned) distance.

20 [Challenging] • Target: Common angles in both units (reference table) • Cross-Lesson: 8.02 Graphing Rational Functions

Consider a circle with its center at the origin $(0, 0)$ and a radius of 5 units. A chord of this circle is perpendicular to the x-axis and intersects the x-axis at the point $(3, 0)$. The endpoints of this chord on the circle are A and B. Determine the length of the chord AB.

The equation of the circle centered at the origin with radius 5 is $x^2 + y^2 = 5^2 = 25$. The chord is perpendicular to the x-axis and intersects it at $(3, 0)$. This means the x-coordinate of the endpoints of the chord, A and B, is 3. Substitute $x = 3$ into the circle's equation to find the corresponding y-coordinates: $3^2 + y^2 = 25$ $9 + y^2 = 25$ $y^2 = 25 - 9$ $y^2 = 16$ $y = \pm 4$ So the endpoints of the chord are $A = (3, 4)$ and $B = (3, -4)$. The length of the chord AB is the distance between these t.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Degrees and Radians Conversion

Convert 150° to exact radians in simplest form.

a. $\frac{5\pi}{6}$ ✓

b. $\frac{2\pi}{3}$

c. $\frac{3\pi}{4}$

d. $\frac{7\pi}{6}$

$150^\circ \times \frac{\pi}{180^\circ} = \frac{150\pi}{180} = \frac{5\pi}{6}$.

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22 [Foundational] • Target: Degrees and Radians Conversion

Convert $\frac{7\pi}{4}$ radians to degrees.

a. 315° ✓

b. 225°

c. 300°

d. 330°

$\frac{7\pi}{4} \times \frac{180^\circ}{\pi} = 7 \times 45^\circ = 315^\circ$.

23 [Foundational] • Target: Degrees and Radians Conversion

Find a positive angle coterminal with $-\frac{2\pi}{3}$ in the interval $[0, 2\pi)$.

a. $\frac{4\pi}{3}$ ✓

b. $\frac{\pi}{3}$

c. $\frac{5\pi}{3}$

d. $\frac{2\pi}{3}$

$-\frac{2\pi}{3} + 2\pi = -\frac{2\pi}{3} + \frac{6\pi}{3} = \frac{4\pi}{3}$.

24 [Foundational] • Target: Arc Length

State the formula for arc length s subtended by a central angle θ (in radians) in a circle of radius r.

a. $s = r\theta$ ✓

b. $s = \frac{1}{2}r^2\theta$

c. $s = 2\pi r\theta$

d. $s = \frac{r}{\theta}$

By the definition of radian measure, the arc length subtended by central angle θ is $s = r\theta$.

25 [Foundational] • Target: Arc Length

Find the arc length intercepted by a central angle of $\frac{2\pi}{3}$ in a circle of radius 9 cm.

a. 6π cm ✓

b. 3π cm

c. 18π cm

d. 12π cm

$s = r\theta = 9 \times \frac{2\pi}{3} = 6\pi$ cm.

26 [Foundational] • Target: Arc Length

An arc of length 10 m subtends a central angle of 2 radians. What is the radius of the circle?

a. 5 m ✓

b. 20 m

c. 10 m

d. 2.5 m

$s = r\theta \Rightarrow 10 = r(2) \Rightarrow r = 5$ m.

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27 [Foundational] • Target: Sector Area

State the formula for the area A of a sector of a circle with radius r and central angle θ (in radians).

a. $A = \frac{1}{2}r^2\theta$ ✓

b. $A = r^2\theta$

c. $A = \frac{1}{2}r\theta$

d. $A = \pi r^2\theta$

The area of a sector proportional to the circle is $\frac{\theta}{2\pi}(\pi r^2) = \frac{1}{2}r^2\theta$.

28 [Foundational] • Target: Sector Area

Find the area of a sector with radius 6 in and central angle $\frac{\pi}{4}$.

a. $\frac{9\pi}{2}$ sq in ✓

b. 9π sq in

c. $\frac{9\pi}{4}$ sq in

d. 18π sq in

$A = \frac{1}{2}r^2\theta = \frac{1}{2}(36)(\frac{\pi}{4}) = 18(\frac{\pi}{4}) = \frac{9\pi}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Sector Area

If a sector of radius 4 cm has an area of 8π cm^2, find its central angle θ.

a. π radians ✓

b. 2π radians

c. $\pi/2$ radians

d. 4π radians

$A = \frac{1}{2}r^2\theta \Rightarrow 8\pi = \frac{1}{2}(16)\theta = 8\theta \Rightarrow \theta = \pi$ radians.

30 [Intermediate] • Target: Linear and Angular Speed

What is the relationship between linear speed v, radius r, and angular speed ω (in rad/time)?

a. $v = r\omega$ ✓

b. $v = \frac{\omega}{r}$

c. $v = r^2\omega$

d. $v = 2\pi r\omega$

Since arc length is $s = r\theta$, dividing by time t gives $\frac{s}{t} = r\frac{\theta}{t} \Rightarrow v = r\omega$.

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31 [Intermediate] • Target: Linear and Angular Speed

A wheel of radius 2 ft rotates at 60 rpm. Find its angular speed in radians per second.

a. 2π rad/s ✓

b. 4π rad/s

c. π rad/s

d. 120π rad/s

$60 \text{ rev/min} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ s}} = 2\pi \text{ rad/s}$.

32 [Intermediate] • Target: Linear and Angular Speed

In the previous question ($r = 2$ ft, $\omega = 2\pi$ rad/s), what is the linear speed of a point on the rim?

a. 4π ft/s ✓

b. 2π ft/s

c. 8π ft/s

d. 120 ft/s

$v = r\omega = 2(2\pi) = 4\pi$ ft/s.

33 [Intermediate] • Target: Linear and Angular Speed

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?

a. **6 units** ✓

b. $6\sqrt{3}$ units

c. 12 units

d. $12\sqrt{3}$ units

We have a circle with radius $r = 6$ and a sector with a central angle of 60 degrees. The chord connecting the endpoints of the arc of this sector forms an isosceles triangle with the two radii. Since the central angle is 60 degrees and the other two angles in the triangle must be equal (because the sides opposite them are radii), each of the other angles is $(180^\circ - 60^\circ)/2 = 60^\circ$. Therefore, the triangle is equilateral. In an equilateral triangle, all sides are equal. Since two sides are radii of.

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34 [Intermediate] • Target: Converting between degrees and radians

In triangle ABC, angle A is 52 degrees. The side opposite angle B (AC) has a length of 10 units, and the side opposite angle C (AB) has a length of 14 units. What is the length of the side opposite angle A (BC)?

a. 10.3

b. **12.1** ✓

c. 14.8

d. 8.5

We can use the Law of Cosines to solve for the unknown side BC. The Law of Cosines states that for any triangle with sides a, b, and c, and opposite angles A, B, and C respectively: $a^2 = b^2 + c^2 - 2bc\cos(A)$. In this problem, we have: side b = AC = 10, side c = AB = 14, and angle A = 52 degrees. We want to find side a = BC. Substituting the values into the formula: $a^2 = 10^2 + 14^2 - 2(10)(14)\cos(52^\circ)$ $a^2 = 100 + 196 - 280\cos(52^\circ)$ $a^2 = 296 - 280(0.61566)$ $a^2 = 296 - 172.3848$.

35 [Intermediate] • Target: Arc length and sector area

In a right-angled triangle ABC, the angle B is 90 degrees. If the length of side AB is 8 units and the length of side BC is 6 units, what is the length of the hypotenuse AC?

a. **10 units** ✓

b. 14 units

c. 48 units

d. 20 units

This is a right-angled triangle, so we can use the Pythagorean theorem: $a^2 + b^2 = c^2$. Here, the sides are $AB = 8$ and $BC = 6$, and the hypotenuse is AC. So, $AC^2 = AB^2 + BC^2$. $AC^2 = 8^2 + 6^2$ $AC^2 = 64 + 36$ $AC^2 = 100$ $AC = \sqrt{100}$ $AC = 10$ units. Common mistakes include adding the sides (8 + 6 = 14) or multiplying them (8 * 6 = 48) instead of applying the Pythagorean theorem.

36 [Intermediate] • Target: Angular and linear speed

In triangle ABC, side $a = 10$ units and side $b = 15$ units. If the angle $C = 60^\circ$, find the length of side c. Round your answer to two decimal places.

a. **13.23** ✓

b. 14.70

c. 12.96

d. 15.52

We can use the Law of Cosines to find the length of side c. The Law of Cosines states that $c^2 = a^2 + b^2 - 2ab\cos C$. Given: $a = 10$, $b = 15$, and $C = 60^\circ$. Substitute the values into the formula: $c^2 = 10^2 + 15^2 - 2(10)(15)\cos 60^\circ$ $c^2 = 100 + 225 - 300 \cdot \frac{1}{2}$ $c^2 = 325 - 150$ $c^2 = 175$ $c = \sqrt{175}$ $c = \sqrt{25 \cdot 7}$ $c = 5\sqrt{7}$ To find the decimal approximation, we calculate: $c \approx 5 \cdot 2.64575$ $c \approx 13.22875$ Rounding to two decimal places.

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37 [Intermediate] • Target: Coterminal angles and reference angles

In triangle ABC, the lengths of sides a, b, and c are opposite to angles A, B, and C respectively. Given that $a = 7$, $b = 10$, and angle $C = 60^\circ$, find the exact length of side c.

a. $\sqrt{79}$

b. 9

c. $\sqrt{91}$ ✓

d. 10

We can use the Law of Cosines to find the length of side c. The Law of Cosines states that $c^2 = a^2 + b^2 - 2ab\cos(C)$. Substituting the given values: $c^2 = 7^2 + 10^2 - 2(7)(10)\cos(60^\circ)$ $c^2 = 49 + 100 - 140(\frac{1}{2})$ $c^2 = 149 - 70$ $c^2 = 79$ Therefore, $c = \sqrt{79}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The radian definition itself • Cross-Lesson: 7.04 Applications: Growth and Decay

Convert $\frac{2\pi}{3}$ radians to degrees.

a. 120° ✓

b. 240°

c. 150°

d. 60°

$\frac{2\pi}{3} \times \frac{180}{\pi} = 120^\circ$.

39 [Challenging] • Target: Combined arc length and sector area problems • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Convert $\frac{5\pi}{4}$ radians to degrees.

a. 225° ✓

b. 255°

c. 210°

d. 135°

$\frac{5\pi}{4} \times \frac{180}{\pi} = 225^\circ$.

40 [Challenging] • Target: Common angles in both units (reference table) • Cross-Lesson: 8.02 Graphing Rational Functions

Convert $\frac{7\pi}{6}$ radians to degrees.

a. 150°

b. 210° ✓

c. 105°

d. 240°

$\frac{7\pi}{6} \times \frac{180}{\pi} = 210^\circ$.

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41 [Challenging] • Target: Satellite and orbital speed applications • Cross-Lesson: 8.03 Introduction to Limits

Convert $\frac{\pi}{3}$ radians to degrees.

a. 300°

b. 60° ✓

c. 90°

d. 30°

$\frac{\pi}{3} \times \frac{180}{\pi} = 60^\circ$.

42 [Challenging] • Target: Negative angles and full-turn arcs • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Convert $\frac{\pi}{4}$ radians to degrees.

a. 30°

b. 75°

c. 45° ✓

d. 315°

$\frac{\pi}{4} \times \frac{180}{\pi} = 45^\circ$.

43 [Challenging] • Target: Degrees and Radians Conversion • Cross-Lesson: 9.02 Geometric Sequences and Series

In a right-angled triangle ABC, where the right angle is at C, if $AC = 10$ and $BC = 24$, find the length of the hypotenuse AB.

a. 25

b. **26** ✓

c. 27

d. 28

We can use the Pythagorean theorem to find the length of the hypotenuse in a right-angled triangle. The theorem states that $a^2 + b^2 = c^2$, where a and b are the lengths of the two shorter sides (legs) and c is the length of the hypotenuse. In this triangle, $AC = 10$ and $BC = 24$. Therefore, we have: $AB^2 = AC^2 + BC^2$ $AB^2 = 10^2 + 24^2$ $AB^2 = 100 + 576$ $AB^2 = 676$ $AB = \sqrt{676}$ $AB = 26$ The length of the hypotenuse AB is 26.

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44 [Challenging] • Target: Arc Length • Cross-Lesson: 9.03 The Binomial Theorem

Two particles, A and B, move in the xy-plane. Their position vectors at time t seconds, for $t \geq 0$, are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$ respectively. The particles start their motion at $t = 0$. Find the minimum distance between particle A and particle B.

a. $\sqrt{26}$ ✓

b. $2\sqrt{13}$

c. $3\sqrt{10}$

d. $4\sqrt{5}$

The position vectors of particle A and particle B at time t are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$. The vector representing the displacement from particle B to particle A at time t is $\mathbf{d}(t) = \mathbf{r}_A(t) - \mathbf{r}_B(t) = (2t - (3 + 5t); (4t - 1) - (1 - 2t)) = (-3t - 3; 6t - 2)$. The distance between the particles at time t is the magnitude of this displacement vector.

45 [Challenging] • Target: Sector Area • Cross-Lesson: 1.01 Functions and Function Notation

Consider a circle with the equation $x^2 + y^2 = 25$. A point $P(x_0, y_0)$ lies on this circle. A tangent line to the circle at point P intersects the x-axis at point A and the y-axis at point B. If the area of the triangle OAB, where O is the origin $(0, 0)$, is minimized, find the value of $x_0^2 + y_0^2$.

a. **25** ✓

b. 50

c. 100

d. 125

The equation of the circle is $x^2 + y^2 = 25$. Let the point of tangency be $P(x_0, y_0)$. The equation of the tangent line to the circle at $P(x_0, y_0)$ is given by $xx_0 + yy_0 = 25$. To find the x-intercept A, we set $y = 0$: $xx_0 = 25 \Rightarrow x = \frac{25}{x_0}$. So, $A = (\frac{25}{x_0}, 0)$. To find the y-intercept B, we set $x = 0$: $yy_0 = 25 \Rightarrow y = \frac{25}{y_0}$. So, $B = (0, \frac{25}{y_0})$. The area of triangle OAB is.

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